# PHARMACOECONOMICS

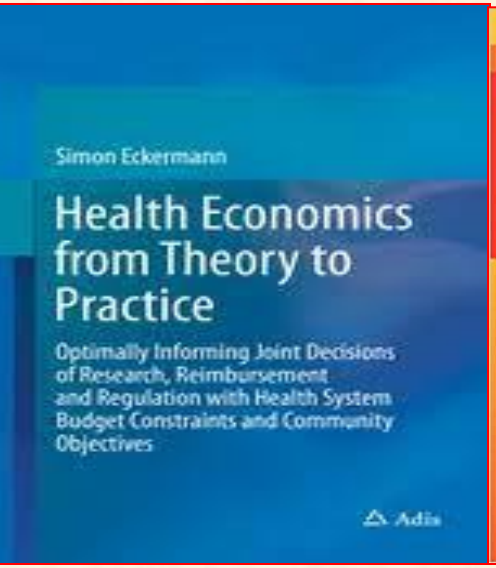


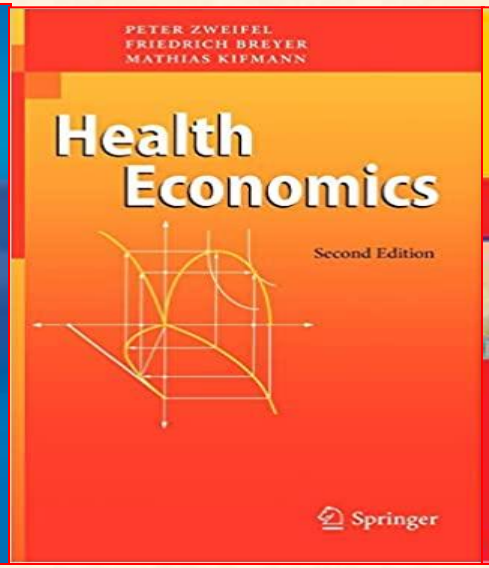


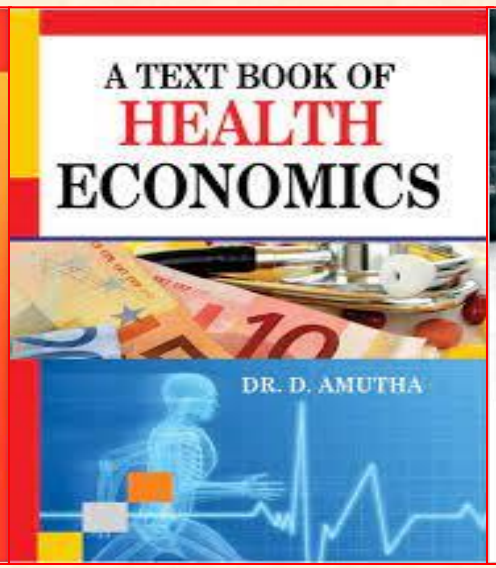

Lecture 1

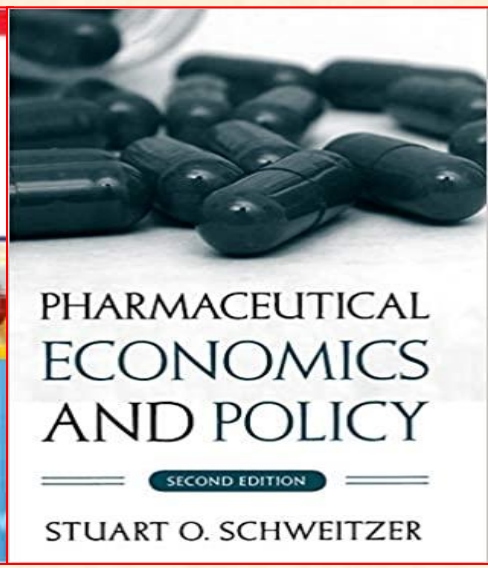
Dr. Sahar Badr Hassan

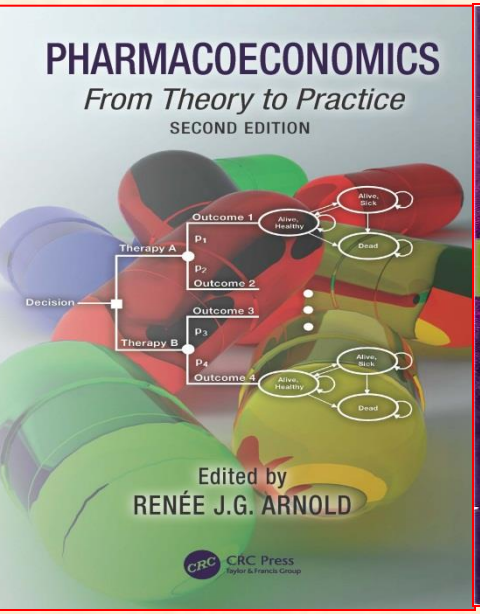
# References

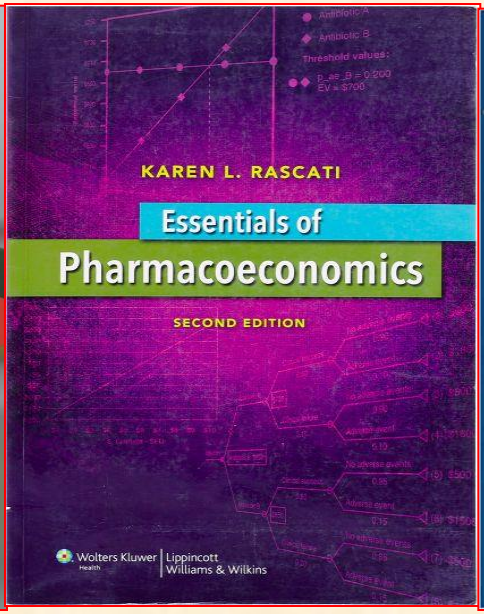


Simon Eckermann
Health Economics from Theory to Practice
Optimally Informing Joint Decisions of Research, Reimbursement and Regulation with Health System Budget Constraints and Community Objectives
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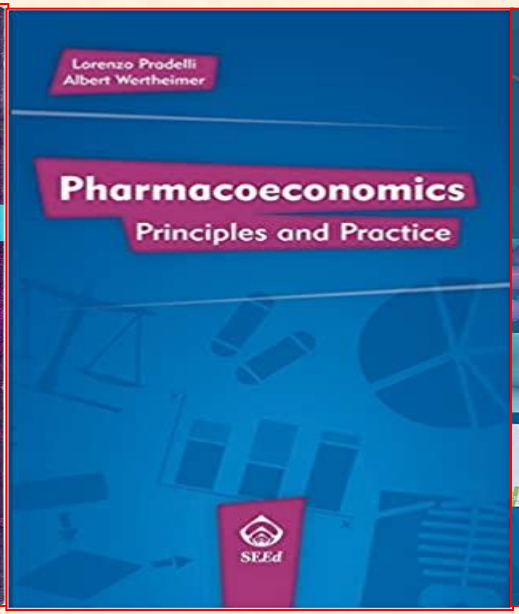

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Health Economics
Second Edition
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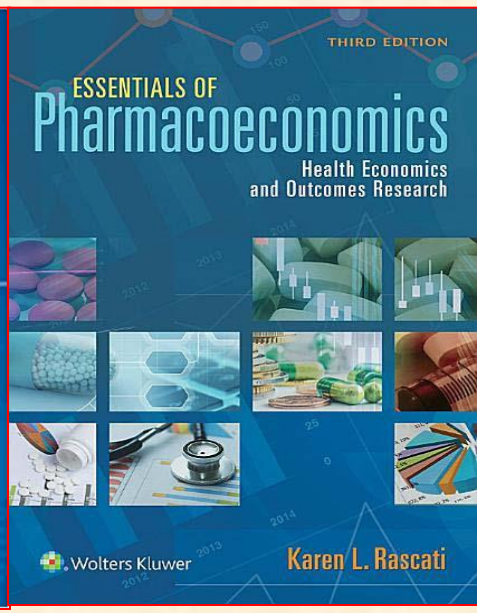

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# Introduction

## What is Economics?

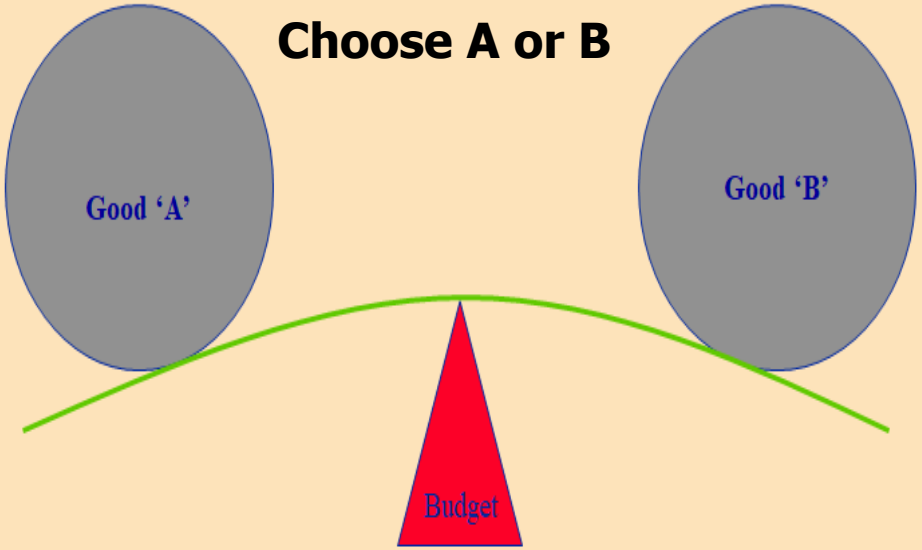
Health Economics and Pharmacoeconomics

What is Pharmacoeconomics?

# Economics is about …

- Limited resources
- Unlimited "wants"
- *Choosing* between which 'wants' we can 'afford' given our resource 'budget'

## Economics is about choice



# Definition of Economics

- There are several definitions of Economics, some of which you can familiarize yourself with include:

- **According to Lionel Robbins,** economics is the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses.

- **Richard Lipsey** defining economics as the study of the use of scarce resources to satisfy unlimited wants.

- Economics is a social science that focuses on the production, distribution, and consumption of goods and services.

- Broadly, economics is the study on how best to allocate our scarce resources.

# Basic Concepts in Economics

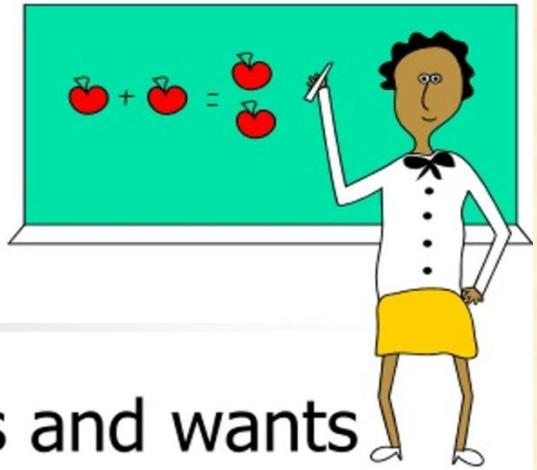
- The basic concepts include; goods and services, scarcity, opportunity cost, choice, economic resources, utility, demand and supply.

- **A good** is a tangible object that is capable of satisfying human wants. Examples of goods are drugs, food, water, cars, clothes, and so on.

- **A service** is an intangible action that is capable of satisfying human want – such services include healthcare services, waste disposal, health sensitization programmes etc.

**Services satisfy our needs as much as goods.**

# Goods and services



- All the things that satisfy needs and wants of the consumers are known as goods and services.

- **Goods** – are things that we can see, feel, and touch. Take home with you, you own it
- **Services** – are things that other people do for you and you pay the for it

| Product | Service |
| --- | --- |
| Products are Tangible | Services are Intangible |
| Products are Imperishable | Services are Perishable |
| Products can be returned | Services can not be returned |
| Products are homogeneous in nature | Services are heterogeneous in nature |
| Quality of a product can be measured | Quality of service is not measurable. |

KINEMASTER

# *Goods and Services satisfy our*

# Needs and Wants

## *Needs …*

The things we ***must have*** in order to survive:

- Water
- Food
- Shelter
- Clothing









## *Wants …*

The things we would **like to have** (but don't really need)

- Fashion clothes
- CDs
- Car
- Holiday







- **Scarcity** is a condition in which it is impossible to satisfy all human wants for goods and services and this forms the central concept in economics.

- In the field of economics, scarcity refers to the limited amount of resources available for man to meet his wants.

- No one can have everything that he or she desires. Therefore, we have to select goods and services we think can give us greatest amount of satisfaction, given our limited resources. This can be well achieved if we arrange our wants in order of importance and allocate our scarce resources to the most important on the list. For example some people may have to choose between going for healthcare and using the available money to pay school fees or house rent.

- Scarcity exists at individual, institutional, community and government levels.

- **Utility** is the amount of satisfaction that a consumer derives from the consumption of the goods or services he/she purchases. It helps to determine how much the consumer is willing to pay.

- **Economic resources** are all natural, human, and material resources which go into the production of goods and services. It is broadly divided into two:

1. Physical resources include land (natural resources) or raw materials and capital.

2. Human resources, which include labour and entrepreneurial ability.

- **Economics**

  ***-The Study of the allocation of scarce resources***

- **Health Economics**

  ***-Economic principles applied to health and healthcare***

  **-Most efficient use of available resources in terms of costs and outcomes**

- **Pharmacoeconomics**

  **-Economic principles applied to drug therapy**

- ***Health*** **refers to a state of the human body and mind.**

- ***Health Care*** **refers to chemicals, devices, and services used by people to improve their health.**

# WHAT IS HEALTH?

. According to World Health Organization's (WHO) constitution health is 'a state of complete physical, mental and social well being and not merely the absence of disease or infirmity'

"Health" in health economic (evaluation) is health status according to some measure.

- Economics
  - Theories to study behavior in allocating scarce resources.
- Health Economics
  - Application of same theories to health and healthcare issues
- Pharmacoeconomics
  - Determination of efficiency in therapeutic purchase and utilization.

## Pharmaceutical Terms:

# EFFICACY, EFFECTIVENESS AND EFFICIENCY




- **Efficiency means maximising benefit form resources used.**

# Definition of Health Economics

- **Health economics** is defined as the application of the theories, concepts, and techniques of economics to the health sector.

- It is concerned with issues like allocation of resources within the various healthcare strategies, quantity and quality of resources used in healthcare delivery, funding of healthcare services, **efficiency** in the use of resources allocated for healthcare and the effects of preventive, curative, and rehabilitative health services on individuals and the society.

# Aim of health economics:

- To quantify overtime the resources used in health service delivery.



- To **organize, allocate and manage** the resources.



# Importance of Health Economics

- Resources in the health sector like other sectors are not enough to satisfy man's health wants.

- The main function of health economics is to apply economic theory to practical problems of rationing the use of resources for effective healthcare services. In response to peoples needs and demands.

- There is **increasing attention on health economics** globally as a result of renewed cost consciousness within the health system and the shift from exclusively humanistic approach to one incorporating an increasing use of managerial techniques and quantitative research methods.

# Concepts in Health Economics

## ➢ Health as an Economic Good

- Health can be seen as an economic good or service. The nature of health is such that it can be seen as a collective good.

- **Collective goods** (or social goods) are defined as the public goods that could be delivered as private goods, but are usually delivered by the government for various reasons, including social policy, and financed from public funds like taxes.

## ➢ Medical Economics

- Often used **synonymously with Health economics**. Medical economics is the branch of economics concerned with the application of economic theory to problem associated typically with pharmaceutical products and medical treatments.

# Health Economic Evaluation

- An economic evaluation is a study that

  compares the **costs** and **benefits** of **two or more alternative interventions**

- SO, the main components are **costs** and **benefits**.




**Costs**
**Direct (fixed, semi-fixed, variable)**
**Indirect**
**Intangible**

**Effects: natural units**
**Utilities: quality of life**
**Benefits: monetary**


THANK
YOU